Model Paper - Database Management Systems

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Q1 (20 marks)

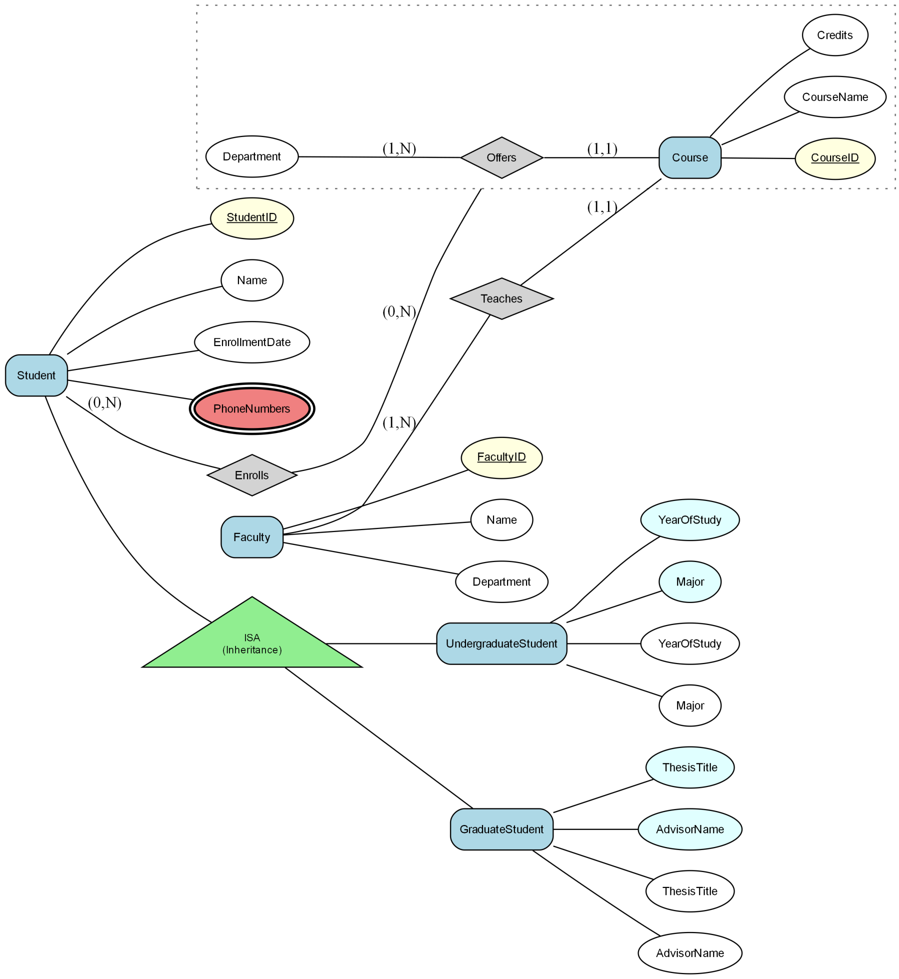
Consider the following Enhanced Entity-Relationship (EER) diagram:

Note: The diagram uses (min, max) cardinality notation where:

- (1,1) indicates one-to-one relationship (each entity participates exactly once)
- (1,N) or (1,*) indicates one-to-many relationship (one entity can relate to many)
- (0,1) indicates optional participation (zero or one)
- (0,N) or (0,*) indicates optional many participation (zero or many)

A university manages information about its courses, students, and faculty. The 'Course' entity has attributes CourseID, CourseName, and Credits. The 'Student' entity consists of StudentID, Name, EnrollmentDate, and PhoneNumbers (a multivalued attribute). Faculty members are captured in the 'Faculty' entity, which includes FacultyID, Name, and Department. Each course is offered by a specific faculty member, establishing a relationship 'Teaches'. Additionally, students enroll in courses through the 'Enrolls' relationship, which includes a composite attribute EnrollmentDetails consisting of EnrollmentDate and Grade. A specialization hierarchy exists where 'Student' further subdivides into 'UndergraduateStudent' with YearOfStudy and Major attributes, and 'GraduateStudent' with ThesisTitle and AdvisorName attributes. This EER diagram includes a situation where a relationship between ['Department', 'Course'] (for example, 'Offers') is treated as a single unit when relating to Student via 'Enrolls'. Model this using an aggregation so that the external relationship is between the external entity and the combined Department-Course-Offers structure, not directly to only one inner entity. A Faculty can be related to many Courses through the 'Teaches' relationship. A Student can be related to many Courses, and a Course can be related to many Students through the 'Enrolls' relationship (Student participation is optional, Course participation is optional)

Figure: EER



a) Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly. (17 marks)

b) What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer. (3 marks)

Q2 (15 marks)

Consider a relation R(A, B, C, D, E) with the following set of functional dependencies F over R: F={{AB->C, B->D, C->EF, DE->B}}.

a) Find all the keys in relation R using the attribute closure method. (4 marks)

b) Is R in 3NF? Give reasons for your conclusion. (3 marks)

c) Is R in BCNF? Give reasons for your conclusion. If R is not in BCNF, convert it to a set of BCNF relations. (8 marks)

Q3 (25 marks)

A retail company is developing a database system to manage its inventory and sales. The database includes tables such as Products, Orders, and Customers. The Products table stores information about items available for sale, including product ID, name, price, and quantity in stock. The Orders table records sales transactions, linking each order to a customer and the purchased products. The Customers table keeps track of customer details, including customer ID, name, and contact information.

This data structure will allow the company to efficiently manage inventory, track sales, and analyze customer purchasing behavior.

a) In Type 2 Driver, JDBC API calls are converted to native Java API calls. Accept or refute the above statement justifying your answer. (2 marks)

b) Code Segment:

```java
String sql = "SELECT * FROM employees WHERE department = ?";
PreparedStatement pstmt = connection.prepareStatement(sql);
pstmt.setString(1, "IT");
ResultSet rs = pstmt.executeQuery();
```

Which type of JDBC statement is used in the code segment shown above?
Briefly explain when this type of statement will be used. (2 marks)

c) Briefly explain the options available for transferring data between two sources such as between tables and servers. (2 marks)

d) Briefly explain how authentication and authorization are achieved in SQL Server. (3 marks)

e) An organization is establishing role-based access control for its database system. Sarah is the senior database administrator responsible for overall database administration. Emily is a junior database administrator responsible for managing the operational database. Nathan is a database developer responsible for creating and maintaining database objects. Michael is a data entry operator responsible for entering and updating transaction data.

i. Write a T-SQL statement to create a login to Sarah with windows authentication. (2 marks)

ii. Provi
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Q4 (40 marks)

Consider the following schema of a database designed for a hospital: Patient (patientId: int, name: varchar(100), address: varchar(150), dob: date) Doctor (doctorId: int, name: varchar(100), specialization: varchar(50), phone: varchar(15)) Appointment (appointmentId: int, appointmentDate: date, patientId: int, doctorId: int) Fee (feeId: int, amount: real, status: varchar(20), appointmentId: int). The 'Patient' table stores the details of patients, the 'Doctor' table keeps records of doctors and their specialties, the 'Appointment' table records the appointments made by patients with doctors, and the 'Fee' table contains details on the fees associated with each appointment. The 'Doctor' table holds details about doctors, including unique doctor ID, name, specialization, and contact information. The

'Appointment' table manages appointment records, tracking scheduled appointments between patients and doctors with dates and times.

a) Write SQL Queries to perform the following:

i. Find the name and phone number of the doctor who has a specialization in 'Pediatrics'. (4 marks)

ii. Find

(6 marks)

b) b) Create a function to calculate the total fee amount for a given patient. This function should account for fee with a late penalty of 10% applied to those marked as 'overdue' payment status. (11 marks)

c) c) Create a trigger that automatically updates the 'amount' column in the 'Fee' table whenever a new fee record is added, or an existing fee record is updated. The 'amount' column should contain the total fee amount for each patient. Ensure the trigger adjusts the 'amount' column accurately to reflect any partial payments made by patient. The patient pays only 50% of the total fee amount as an initial payment when the payment status is marked as 'Partial'. (12 marks)